Отчет по практике 2 модуль 1 раздел 1

Шаг 1

Создайте на сервере для 1 практики ключ ssh при помощи программы ssh-keygen

Ssh key-gen

```
eltex-pg2-v16@eltex-2026-spring-17:42> ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/eltex-pg2-v16/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/eltex-pg2-v16/.ssh/id_rsa
Your public key has been saved in /home/eltex-pg2-v16/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:kIWnkNlBJ3Yigmt/s83uZ0KkR3cp9cwIxfIYuzHr9OY eltex-pg2-v16@eltex-2026-spring
The key's randomart image is:
+---[RSA 3072]----+
| .. .=*o+ o.     |
|.  .+ooB.+ o     |
| .   .oo  O *    |
|..    .+ B = +   |
|. .   + S B      |
|   . + o +       |
|    . B o .      |
|     . + + o     |
|      oo+ oE     |
+----[SHA256]-----+
eltex-pg2-v16@eltex-2026-spring-17:48> cat .ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCCbOKxPkheBlixeRweJEmsjCz90J6a3lTl14uzxCS0Sh1loJeo8BUpa
gD2Vey8vrKa/p7vUxobDTErcc7eYi+Ci8xCwafZJ9Ymccf2haLCqBAta27PS8B1G72EJxN/nNO1iQASld6gJyyQphNNcw
HEKHmVCgJQYCjQhhAQDxwN1QgFmEIkvSg7oyzil8GYZQJgw/RsovIShoU5MOpaDSSikDU7wCs5HqiVmRLTd3Eu5JKmi9N
ySx2DrFrmK11GgFVDMMVt3jpnoIU6E4RGa4OrcJGIt9eGvrMVA8qgaaeKkTv3/kg8aaX23gqMEwV9jYFLWCbw82Wq+7SX
pGNJRWGRAOABJfGgnwPmrCeRyd71Y0GiHJXFtRXBviIopnD9UwzfOx/oRQk/sAc0PwDz5nXRNPBGC4RzevDGA7IYtRg0i
RL5F78/zik1vrUCX5FJ08wc1cj+oaXGNetAHkRfonA0xEjltS9Pgm+1639Nbs642Zz7u1jbTA5ytDk0w3ACXcM= eltex
-pg2-v16@eltex-2026-spring
eltex-pg2-v16@eltex-2026-spring-17:48>
```

Шаг 2

Скопируйте созданный ключ на сервер для 2 практики для пользователя root при помощи программы ssh-copy-id

ssh-copy-id -i ~/.ssh/id_rsa.pub root@172.16.9.216

Шаг 3

Подключитесь к серверу для 2 практики под пользователем root и сравните содержимое файла, а так же права доступа для каждого из файлов

cat ~/.ssh/*.pub

```
eltex-pg2-v16@eltex-2026-spring-18:32> cat ~/.ssh/*.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCCbOKxPkheBlixeRweJEmsjCz90J6a3lTl14uzxCS0
Sh1loJeo8BUpagD2Vey8vrKa/p7vUxobDTErcc7eYi+Ci8xCwafZJ9Ymccf2haLCqBAta27PS8B1G72E
JxN/nNO1iQASld6gJyyQphNNcwHEKHmVCgJQYCjQhhAQDxwN1QgFmEIkvSg7oyzil8GYZQJgw/RsovIS
hoU5MOpaDSSikDU7wCs5HqiVmRLTd3Eu5JKmi9NySx2DrFrmK11GgFVDMMVt3jpnoIU6E4RGa4OrcJGI
t9eGvrMVA8qgaaeKkTv3/kg8aaX23gqMEwV9jYFLWCbw82Wq+7SXpGNJRWGRAOABJfGgnwPmrCeRyd71
Y0GiHJXFtRXBviIopnD9UwzfOx/oRQk/sAc0PwDz5nXRNPBGC4RzevDGA7IYtRg0iRL5F78/zik1vrUC
X5FJ08wc1cj+oaXGNetAHkRfonA0xEjltS9Pgm+1639Nbs642Zz7u1jbTA5ytDk0w3ACXcM= eltex-p
g2-v16@eltex-2026-spring
```

ssh root@172.16.9.216

```
root@eltex-practice2-pg2-v16:~# cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCCbOKxPkheBlixeRweJEmsjCz90J6a3lTl14uzxCS0
Sh1loJeo8BUpagD2Vey8vrKa/p7vUxobDTErcc7eYi+Ci8xCwafZJ9Ymccf2haLCqBAta27PS8B1G72E
JxN/nNO1iQASld6gJyyQphNNcwHEKHmVCgJQYCjQhhAQDxwN1QgFmEIkvSg7oyzil8GYZQJgw/RsovIS
hoU5MOpaDSSikDU7wCs5HqiVmRLTd3Eu5JKmi9NySx2DrFrmK11GgFVDMMVt3jpnoIU6E4RGa4OrcJGI
t9eGvrMVA8qgaaeKkTv3/kg8aaX23gqMEwV9jYFLWCbw82Wq+7SXpGNJRWGRAOABJfGgnwPmrCeRyd71
Y0GiHJXFtRXBviIopnD9UwzfOx/oRQk/sAc0PwDz5nXRNPBGC4RzevDGA7IYtRg0iRL5F78/zik1vrUC
X5FJ08wc1cj+oaXGNetAHkRfonA0xEjltS9Pgm+1639Nbs642Zz7u1jbTA5ytDk0w3ACXcM= eltex-p
g2-v16@eltex-2026-spring
```

```
eltex-pg2-v16@eltex-2026-spring-13:58> ls -la ~/.ssh/
total 24
drwx------  2 eltex-pg2-v16 eltex-pg2-v16 4096 мар 12 17:59 .
drwxr-xr-x 17 eltex-pg2-v16 eltex-pg2-v16 4096 мар 12 18:41 ..
-rw-------  1 eltex-pg2-v16 eltex-pg2-v16 2622 мар 12 17:48 id_rsa
-rw-r--r--  1 eltex-pg2-v16 eltex-pg2-v16  585 мар 12 17:48 id_rsa.pub
-rw-------  1 eltex-pg2-v16 eltex-pg2-v16  978 мар 12 17:20 known_hosts
-rw-r--r--  1 eltex-pg2-v16 eltex-pg2-v16  142 мар 12 17:20 known_hosts.old
eltex-pg2-v16@eltex-2026-spring-14:16>
```

```
ls -la ~/.ssh/
total 12
drwx------ 2 root root 4096 Mar 12  2025 .
drwx------ 4 root root 4096 Mar 13 06:55 ..
-rw------- 1 root root  585 Mar 12 10:59 authorized_keys
root@eltex-practice2-pg2-v16:~#
```

Шаг 4

Создайте пользователя user1 при помощи команды useradd, укажите необходимость создания домашнего каталога и shell /bin/bash. Создайте пароль пользователю user1

useradd -m -s /bin/bash user1

```
root@eltex-practice2-pg2-v16:~# useradd -m -s /bin/bash user1
root@eltex-practice2-pg2-v16:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg2-v16:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1)
```

Шаг 5

Создайте пользователя user2 и user3 при помощи команды adduser

adduser user2

```
root@eltex-practice2-pg2-v16:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
root@eltex-practice2-pg2-v16:~#
```

adduser user3

```
root@eltex-practice2-pg2-v16:~# adduser user3
info: Adding user `user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user3' (1003) ...
info: Adding new user `user3' (1003) with group `user3 (1003)' ...
info: Creating home directory `/home/user3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user3' to supplemental / extra groups `users' ...
info: Adding user `user3' to group `users' ...
root@eltex-practice2-pg2-v16:~#
```

Шаг 6

Для пользователя user3 смените shell на /usr/sbin/nologin (man usermod), выполните вход под этим пользователем при помощи утилиты su, сначала без дополнительных параметров, затем с явным указанием shell /bin/bash в параметрах su. Выполните logout

Usermod -s /usr/sbin/nologin user3

```
usermod -s /usr/sbin/nologin user3
root@eltex-practice2-pg2-v16:~# grep user3 /etc/passwd
user3:x:1003:1003:,,,:/home/user3:/usr/sbin/nologin
root@eltex-practice2-pg2-v16:~#
```

Su – user3

```
root@eltex-practice2-pg2-v16:~# su - user3
This account is currently not available.
root@eltex-practice2-pg2-v16:~#
```

su -s /bin/bash user3

```
root@eltex-practice2-pg2-v16:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v16:/root$
```

Exit

Шаг 7-8

Создайте новую группу и добавьте её для всех пользователей user* как дополнительную, посмотрите список групп всех пользователей user*

groupadd onegroup

usermod -aG onegroup user1

usermod -aG onegroup user2

usermod -aG onegroup user3

```
root@eltex-practice2-pg2-v16:~# usermod -aG onegroup user1
root@eltex-practice2-pg2-v16:~# usermod -aG onegroup user2
root@eltex-practice2-pg2-v16:~# usermod -aG onegroup user3
```

Id user1

```
root@eltex-practice2-pg2-v16:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1),1004(onegroup)
root@eltex-practice2-pg2-v16:~#
```

Id user2

```
root@eltex-practice2-pg2-v16:~# id user2
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users),1004(onegroup)
root@eltex-practice2-pg2-v16:~#
```

Id user3

```
root@eltex-practice2-pg2-v16:~# id user3
uid=1003(user3) gid=1003(user3) groups=1003(user3),100(users),1004(onegroup)
root@eltex-practice2-pg2-v16:~#
```

Шаг 9

Создайте каталог /opt/share и назначьте группу из предыдущего пункта его владельцем, установите на этот каталог бит SGID, права для группы rwx.

mkdir -p /opt/share

chgrp onegroup /opt/share

chmod g+s,g+rwx /opt/share

```
root@eltex-practice2-pg2-v16:~# chmod g+s,g+rwx /opt/share
root@eltex-practice2-pg2-v16:~# echo "chmod g+s,g+rwx /opt/share" >> practice2_e
ltex-pg2-v16_part1.log
root@eltex-practice2-pg2-v16:~# ls -ld /opt/share
drwxrwsr-x 2 root onegroup 4096 Mar 13 08:40 /opt/share
root@eltex-practice2-pg2-v16:~#
```

Шаг 10

Для user1 задайте перманентно umask, снимающий право чтения для «прочих»

echo "umask 027" >> /home/user1/.bashrc

```
root@eltex-practice2-pg2-v16:~# su - user1
user1@eltex-practice2-pg2-v16:~$ umask
0027
user1@eltex-practice2-pg2-v16:~$
```

Шаг 11

Создайте каждым из пользователей новые файлы в каталоге /opt/share, удалите файлы созданные другими пользователями

su - user1

touch /opt/share/file_user1.txt

exit

```
root@eltex-practice2-pg2-v16:~# su - user1
user1@eltex-practice2-pg2-v16:~$ touch /opt/share/file_user1.txt
```

su - user2

touch /opt/share/file_user2.txt

exit

su -s /bin/bash user3

touch /opt/share/file_user3.txt

exit

```
root@eltex-practice2-pg2-v16:~# su - user2
user2@eltex-practice2-pg2-v16:~$ touch /opt/share/file_user2.txt
user2@eltex-practice2-pg2-v16:~$ exit
logout
root@eltex-practice2-pg2-v16:~# su - user3
This account is currently not available.
root@eltex-practice2-pg2-v16:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v16:/root$ touch /opt/share/file_user3.txt
user3@eltex-practice2-pg2-v16:/root$ exit
exit
```

```
root@eltex-practice2-pg2-v16:~# ls -la /opt/share/
total 8
drwxrwsr-x 2 root  onegroup 4096 Mar 13 09:31 .
drwxr-xr-x 3 root  root     4096 Mar 13 08:40 ..
-rw-r----- 1 user1 onegroup    0 Mar 13 09:24 file_user1.txt
-rw-rw-r-- 1 user2 onegroup    0 Mar 13 09:29 file_user2.txt
-rw-rw-r-- 1 user3 onegroup    0 Mar 13 09:31 file_user3.txt
root@eltex-practice2-pg2-v16:~# vi practice2_eltex-pg2-v16_part1.log
root@eltex-practice2-pg2-v16:~#
```

//Попытка удалить файл созданый другим пользователем

Su – user1

Rm /opt/share/file_user2.txt

exit

```
user1@eltex-practice2-pg2-v16:~$ rm /opt/share/file_user2.txt
user1@eltex-practice2-pg2-v16:~$
```

```
root@eltex-practice2-pg2-v16:~# ls -la /opt/share/
total 8
drwxrwsr-x 2 root  onegroup 4096 Mar 13 09:41 .
drwxr-xr-x 3 root  root     4096 Mar 13 08:40 ..
-rw-r----- 1 user1 onegroup    0 Mar 13 09:24 file_user1.txt
-rw-rw-r-- 1 user3 onegroup    0 Mar 13 09:31 file_user3.txt
root@eltex-practice2-pg2-v16:~#
```

Шаг 12

Повторите предыдущий пункт, предварительно установив sticky bit на каталоге /opt/share

chmod +t /opt/share

```
root@eltex-practice2-pg2-v16:~# ls -ld /opt/share
drwxrwsr-t 2 root onegroup 4096 Mar 13 09:51 /opt/share
root@eltex-practice2-pg2-v16:~#
```

su - user1 -c "touch /opt/share/file2_user1.txt"

su - user2 -c "touch /opt/share/file2_user2.txt"

su -s /bin/bash user3 -c "touch /opt/share/file2_user3.txt"

```
root@eltex-practice2-pg2-v16:~# ls -la /opt/share
total 8
drwxrwsr-t 2 root  onegroup 4096 Mar 13 10:08 .
drwxr-xr-x 3 root  root     4096 Mar 13 08:40 ..
-rw-rw-r-- 1 user1 onegroup    0 Mar 13 10:01 file2_user1.txt
-rw-rw-r-- 1 user2 onegroup    0 Mar 13 10:08 file2_user2.txt
-rw-rw-r-- 1 user3 onegroup    0 Mar 13 10:02 file2_user3.txt
root@eltex-practice2-pg2-v16:~#
```

su - user1 -c "rm /opt/share/file2_user2.txt"

```
root@eltex-practice2-pg2-v16:~# su - user1 -c "rm /opt/share/file2_user2.txt"
rm: cannot remove '/opt/share/file2_user2.txt': Operation not permitted
root@eltex-practice2-pg2-v16:~#
```

Шаг 13

Разрешите user1 выполнять привилегированную команду dmesg при помощи команды sudo, а user2 – при помощи скрипта на языке bash с установленным флагом SUID

Visudo

//добавляем строчку

user1 ALL=(ALL) NOPASSWD: /usr/bin/dmesg

```
root@eltex-practice2-pg2-v16: ~
  GNU nano 7.2                          /etc/sudoers.tmp *
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
user1 ALL=(ALL) NOPASSWD: /usr/bin/dmesg

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL
```

Nano /usr/local/bin/dmesg_script.sh

Вводим:

#!/bin/bash

/usr/bin/dmesg

chown root:root /usr/local/bin/dmesg_script.sh

chmod 755 /usr/local/bin/dmesg_script.sh

chmod u+s /usr/local/bin/dmesg_script.sh

ls -la /usr/local/bin/dmesg_script.sh

```
root@eltex-practice2-pg2-v16:~# nano /usr/local/bin/dmesg_script.sh
root@eltex-practice2-pg2-v16:~# chown root:root /usr/local/bin/dmesg_script.sh
root@eltex-practice2-pg2-v16:~# chmod 755 /usr/local/bin/dmesg_script.sh
root@eltex-practice2-pg2-v16:~# chmod u+s /usr/local/bin/dmesg_script.sh
root@eltex-practice2-pg2-v16:~# ls -la /usr/local/bin/dmesg_script.sh
-rwsr-xr-x 1 root root 27 Mar 13 11:22 /usr/local/bin/dmesg_script.sh
root@eltex-practice2-pg2-v16:~#
```

Шаг 14

Для всех пользователей user* задайте время действия пароля – 10 дней.

chage -M 10 user1

chage -M 10 user2

chage -M 10 user3

```
root@eltex-practice2-pg2-v16:~# chage -l user1
Last password change                                    : Mar 13, 2026
Password expires                                        : Mar 23, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
root@eltex-practice2-pg2-v16:~# chage -l user2
Last password change                                    : Mar 13, 2026
Password expires                                        : Mar 23, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
root@eltex-practice2-pg2-v16:~# chage -l user3
Last password change                                    : Mar 13, 2026
Password expires                                        : Mar 23, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
```

Шаг 15

Отредактируйте файл /etc/motd, вписав туда свое имя и фамилию

nano /etc/motd

```
root@eltex-practice2-pg2-v16: ~
  GNU nano 7.2                    /etc/motd
Алина Дережинцева
```

шаг 16

Создайте копию содержимого каталога /etc в каталог /root/etc_backup при помощи программы rcync

rsync -av /etc/ /root/etc_backup/

```
root@eltex-practice2-pg2-v16:~# ls -la /root/etc_backup/
total 940
drwxr-xr-x 109 root root       4096 Mar 13 12:01 .
drwx------   6 root root       4096 Mar 13 12:20 ..
-rw-r--r--   1 root root       3444 Jul  5  2023 adduser.conf
drwxr-xr-x   2 root root       4096 Mar 13 09:31 alternatives
drwxr-xr-x   2 root root       4096 Feb 16  2025 apparmor
drwxr-xr-x   9 root root       4096 Feb 16  2025 apparmor.d
drwxr-xr-x   3 root root       4096 Mar 13 08:08 apport
drwxr-xr-x   9 root root       4096 Mar 12  2025 apt
-rw-r--r--   1 root root       2319 Mar 31  2024 bash.bashrc
-rw-r--r--   1 root root         45 Feb 16  2025 bash_completion
drwxr-xr-x   2 root root       4096 Mar 13 09:05 bash_completion.d
-rw-r--r--   1 root root        367 Aug  2  2022 bindresvport.blacklist
drwxr-xr-x   2 root root       4096 Oct 17  2024 binfmt.d
drwxr-xr-x   2 root root       4096 Feb 16  2025 byobu
drwxr-xr-x   3 root root       4096 Feb 16  2025 ca-certificates
-rw-r--r--   1 root root       6288 Feb 16  2025 ca-certificates.conf
drwxr-xr-x   5 root root       4096 Mar 13 09:29 cloud
drwxr-xr-x   2 root root       4096 Mar 12  2025 console-setup
drwx------   2 root root       4096 Oct 17  2024 credstore
```

```
root@eltex-practice2-pg2-v16:~# find /etc -type f | wc -l
859
root@eltex-practice2-pg2-v16:~# find /root/etc_backup -type f | wc -l
859
root@eltex-practice2-pg2-v16:~#
```

Шаг 17

Заархивируйте содержимое каталога /root/etc_backup архиватором tar, используйте алгоритмы сжатия gzip, bzip2, 7zip, сравните размеры полученных файлов

tar -czvf /root/etc_backup.gz /root/etc_backup

```
root@eltex-practice2-pg2-v16:~# tar -czvf /root/etc_backup.gz /root/etc_backup
tar: Removing leading `/' from member names
/root/etc_backup/
/root/etc_backup/rc4.d/
/root/etc_backup/rc4.d/S01uuidd
/root/etc_backup/rc4.d/S01grub-common
/root/etc_backup/rc4.d/S01rsync
/root/etc_backup/rc4.d/S01plymouth
/root/etc_backup/rc4.d/S01dbus
/root/etc_backup/rc4.d/S01sysstat
/root/etc_backup/rc4.d/S01console-setup.sh
/root/etc_backup/rc4.d/S01open-vm-tools
/root/etc_backup/rc4.d/S01ssh
/root/etc_backup/rc4.d/S01qemu-guest-agent
/root/etc_backup/rc4.d/S01cron
/root/etc_backup/rc4.d/S01unattended-upgrades
/root/etc_backup/rc4.d/S01apport
/root/etc_backup/inputrc
/root/etc_backup/credstore/
/root/etc_backup/issue.net
/root/etc_backup/gai.conf
/root/etc_backup/machine-id
```

apt-get install bzip2 -y

tar -cjvf /root/etc_backup.bz2 /root/etc_backup

```
root@eltex-practice2-pg2-v16:~# tar -cjvf /root/etc_backup.bz2 /root/etc_backup
tar: Removing leading `/' from member names
/root/etc_backup/
/root/etc_backup/rc4.d/
/root/etc_backup/rc4.d/S01uuidd
/root/etc_backup/rc4.d/S01grub-common
/root/etc_backup/rc4.d/S01rsync
/root/etc_backup/rc4.d/S01plymouth
/root/etc_backup/rc4.d/S01dbus
/root/etc_backup/rc4.d/S01sysstat
/root/etc_backup/rc4.d/S01console-setup.sh
/root/etc_backup/rc4.d/S01open-vm-tools
/root/etc_backup/rc4.d/S01ssh
/root/etc_backup/rc4.d/S01qemu-guest-agent
/root/etc_backup/rc4.d/S01cron
/root/etc_backup/rc4.d/S01unattended-upgrades
/root/etc_backup/rc4.d/S01apport
/root/etc_backup/inputrc
/root/etc_backup/credstore/
/root/etc_backup/issue.net
/root/etc_backup/gai.conf
```

apt-get install p7zip-full -y

7z a /root/etc_backup.7z /root/etc_backup

```
root@eltex-practice2-pg2-v16:~# 7z a /root/etc_backup.7z /root/etc_backup
Command '7z' not found, but can be installed with:
apt install 7zip        # version 23.01+dfsg-7, or
apt install p7zip-full  # version 16.02+dfsg-8
root@eltex-practice2-pg2-v16:~# apt-get install p7zip-full -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  7zip
Suggested packages:
  7zip-standalone 7zip-rar
The following NEW packages will be installed:
  7zip p7zip-full
0 upgraded, 2 newly installed, 0 to remove and 87 not upgraded.
Need to get 1,849 kB of archives.
After this operation, 6,318 kB of additional disk space will be used.
Get:1 http://ru.archive.ubuntu.com/ubuntu noble/universe amd64 7zip amd64 23.01+
dfsg-11 [1,846 kB]
Get:2 http://ru.archive.ubuntu.com/ubuntu noble/universe amd64 p7zip-full all 16
```

ls -lh /root/etc_backup.*

```
root@eltex-practice2-pg2-v16:~# ls -lh /root/etc_backup.*
-rw-r--r-- 1 root root 4.5M Mar 13 12:31 /root/etc_backup.7z
-rw-r--r-- 1 root root 553K Mar 13 12:44 /root/etc_backup.bz2
-rw-r--r-- 1 root root 584K Mar 13 12:27 /root/etc_backup.gz
```

```
root@eltex-practice2-pg2-v16:~# ls
etc_backup     etc_backup.bz2  practice2_eltex-pg2-v16_part1.log
etc_backup.7z  etc_backup.gz
root@eltex-practice2-pg2-v16:~#
```

Шаг 18

Отредактируйте файл /etc/motd, вписав туда текущую дату и время, синхронизируйте каталог /root/etc_backup с каталогом /etc при помощи rsync, добавьте файл motd в архив, сжатый gzip

Nano /etc/motd

```
Алина Дережинцева
13.03.2026 19:51
```

rsync -av /etc/ /root/etc_backup/

```
root@eltex-practice2-pg2-v16:~# rsync -av /etc/ /root/etc_backup/
sending incremental file list
./
motd

sent 70,008 bytes  received 286 bytes  140,588.00 bytes/sec
total size is 2,366,900  speedup is 33.67
root@eltex-practice2-pg2-v16:~#
```

time rsync -av /etc/ /root/etc_backup/

```
root@eltex-practice2-pg2-v16:~# time rsync -av /etc/ /root/etc_backup/
sending incremental file list

sent 69,907 bytes  received 260 bytes  140,334.00 bytes/sec
total size is 2,366,900  speedup is 33.73

real    0m0.050s
user    0m0.006s
sys     0m0.006s
root@eltex-practice2-pg2-v16:~#
```

```
root@eltex-practice2-pg2-v16:~# ls -la /etc/motd /root/etc_backup/motd
-rw-r--r-- 1 root root 51 Mar 13 12:51 /etc/motd
-rw-r--r-- 1 root root 51 Mar 13 12:51 /root/etc_backup/motd
root@eltex-practice2-pg2-v16:~#
```

rm /root/etc_backup.gz

tar -czvf /root/etc_backup.gz -C / root/etc_backup

Шаг 19

Сравните содержимое архива, упакованного bzip2 с сорержимым каталога /root/etc_backup

//нужно создать временную директорию куда распакуем архив bzip2

mkdir -p /root/etc_backup_bzip2_test

//распаковать архив

tar -xjvf /root/etc_backup.bz2 -C /root/etc_backup_bzip2_test

//сравнить с оригиналом

diff -rq /root/etc_backup_bzip2_test/root/etc_backup /root/etc_backup

```
root@eltex-practice2-pg2-v16:~# diff -rq /root/etc_backup_bzip2_test/root/etc_backup /roo
t/etc_backup
Files /root/etc_backup_bzip2_test/root/etc_backup/motd and /root/etc_backup/motd differ
diff: /root/etc_backup_bzip2_test/root/etc_backup/mtab: No such file or directory
diff: /root/etc_backup/mtab: No such file or directory
diff: /root/etc_backup_bzip2_test/root/etc_backup/os-release: No such file or directory
diff: /root/etc_backup/os-release: No such file or directory
diff: /root/etc_backup_bzip2_test/root/etc_backup/resolv.conf: No such file or directory
diff: /root/etc_backup/resolv.conf: No such file or directory
root@eltex-practice2-pg2-v16:~#
```

Шаг 20

Распакуйте архивы etc_backup, упакованные gzip и 7zip в каталоги /root/etc_backup_gzip и /root/etc_backup_7zip, сравните программой diff файлы motd в этих каталогах.

//Создать директории для распаковки

mkdir -p /root/etc_backup_gzip

mkdir -p /root/etc_backup_7zip

//распаковать архивы

tar -xzvf /root/etc_backup.gz -C /root/etc_backup_gzip

7z x /root/etc_backup.7z -o/root/etc_backup_7zip

//Сравнить файлы

diff -u /root/etc_backup_gzip/root/etc_backup/motd /root/etc_backup_7zip/etc_backup/motd

```
root@eltex-practice2-pg2-v16:~# diff -u /root/etc_backup_gzip/root/etc_backup/motd /root/etc_backup_7zip/etc_backup/motd
--- /root/etc_backup_gzip/root/etc_backup/motd  2026-03-13 12:51:48.000000000 +0000
+++ /root/etc_backup_7zip/etc_backup/motd       2026-03-13 12:01:23.304317300 +0000
@@ -1,2 +1 @@
 Алина Дережинцева
-13.03.2026 19:51
root@eltex-practice2-pg2-v16:~#
```